

Standard Operating Procedure

Fire Alarm Interface & Life-Safety Interlock for Access-Controlled Doors

Document No.	SOP-LV-014
Related Documents	SOP-LV-006, SOP-LV-012
Revision	1.0
Effective Date	2026-07-23
Review Cycle	Annual
Applies To	Field Technicians, Installers
Jurisdiction	Ontario (see Section 9 for Manitoba)
Owner	Apex – Operations
Approved By	_____

THIS IS A LIFE-SAFETY AND CODE COMPLIANCE PROCEDURE.

The requirements below are drawn directly from Article 3.4.6.16 of the Ontario Building Code (Door Release Hardware). This is not a general best-practice guide — these are code-mandated minimums. Do not deviate from this procedure without written sign-off from the Site Supervisor and confirmation against the current OBC text, since code text is periodically amended. A door that fails to release on fire alarm activation is a life-safety failure, not just a system fault.

1. Purpose

This procedure defines the mandatory interlock requirements between an access-controlled door's locking hardware and the building's fire alarm system, ensuring the door releases for egress under all the conditions required by the Ontario Building Code.

2. Scope

Applies to any door on an access-controlled egress path (a door in the path of exit travel) where an electromagnetic lock or other electrified locking hardware is installed. Does not apply to doors where the egress side is always mechanically free to open — the OBC interlock requirements exist specifically because an electromagnetic lock secures both sides of the door.

3. Mandatory Release Conditions — OBC Article 3.4.6.16

Per Article 3.4.6.16 of the Ontario Building Code, an electromagnetic (or similar) locking device on a door in an access to exit is only permitted where the building has a conforming fire alarm system AND the locking device is installed as an ancillary device to that fire alarm system, releasing immediately upon ALL of the following:

- Activation of the fire alarm signal (or the alert signal, where a two-stage fire alarm system is installed — with an exception for two-stage systems in care, care and treatment, or detention occupancies, which release on the alarm signal, not the alert signal)
- Loss of power to the fire alarm control panel, or loss of power controlling the electromagnetic locking mechanism and its associated auxiliary controls
- Operation of a manually operated switch, readily accessible only to authorized personnel, located near the main entrance of the building or in the central alarm and control facility
- A fault being detected in the electrical circuit between the fire alarm control panel and the locking device's controller
- Operation of a manual pull station for the fire alarm system located on the wall not more than 600 mm from the door

4. Required Signage

A permanently mounted, legible sign reading “EMERGENCY EXIT UNLOCKED BY FIRE ALARM” is required on the door, per OBC Article 3.4.6.16. Lettering must be at least 25 mm high with the required stroke width per the code text — confirm exact sign specification against the current OBC before ordering signage, and do not substitute wording.

5. What This Means for Installation

- The lock's release must be wired as a hard interlock to the fire alarm system — not dependent on the access control panel, EntraPass software, network connectivity, or any device that could itself fail, lose power, or lose communication without also releasing the lock.
- Confirm the connection method with the fire alarm contractor/system integrator for the building — the exact interface point (relay output on the fire alarm panel, dedicated door-release circuit, etc.) is specific to that fire alarm system and is not something to guess at or wire speculatively.
- The access control system (Kantech/EntraPass) may control the lock for normal access purposes, but the fire alarm interlock must remain a physically separate release path that functions even if the access control system is fully powered down or malfunctioning.
- Test the fault-detection requirement specifically — confirm the lock releases if the wiring between the fire panel and the lock controller is cut or opened, not just when the fire alarm actively signals. This is frequently missed because it requires an active fault test, not just an alarm activation test.

6. Coordination Requirements

- Do not perform fire alarm interlock work without coordinating with the building's fire alarm contractor or the authority having jurisdiction (AHJ), as applicable — this is not a low-voltage-only scope of work.
- Where Apex is not the fire alarm contractor for the site, confirm in writing which company is responsible for the fire-alarm-side interface before wiring anything on that side of the interlock.
- Any change to a door's locking hardware or its fire alarm interlock on an existing installation should be flagged to the site's fire alarm verification/inspection records — this work can trigger a requirement for re-verification of the fire alarm system by a qualified fire alarm technician.

7. Testing & Verification

- Test full fire alarm activation — confirm the door releases immediately and stays released/unlocked.
- Test the manual pull station within 600 mm of the door, if present — confirm it independently releases the lock.
- Test loss of power to the fire alarm control panel and to the lock's auxiliary controls — confirm fail-safe release in both cases.
- Test the authorized-personnel manual override switch near the main entrance or central alarm facility.
- Test for fault detection: open the circuit between the fire alarm panel and the lock controller and confirm the lock releases.
- Confirm the door does not re-lock until the fire alarm panel is reset — the lock must not re-secure automatically once the alarm condition clears on its own.
- Confirm required signage is installed, legible, and correctly worded before considering the door complete.

8. Documentation

Record the interlock test results (Section 7) on the site's commissioning documentation, including date, technician, and confirmation of each release condition tested. This record should be available for the site's fire alarm verification file, since it may be requested by the AHJ or the fire alarm verification technician.

9. Manitoba Note

For work performed in Manitoba, the equivalent requirements are found in the Manitoba Building Code and the applicable fire code — do not assume the Ontario Building Code article numbers or exact release conditions apply unchanged. Confirm the current Manitoba requirements with the Site Supervisor or a qualified fire alarm contractor before performing this work in that jurisdiction.

10. References

- Ontario Building Code, Division B, Article 3.4.6.16 — Door Release Hardware
- Ontario Building Code, Division B, Article 3.2.4 — Fire Alarm and Detection Systems (referenced by 3.4.6.16)
- Ontario Fire Code
- UL 294 — Access Control System Units
- SOP-LV-006 — Utility Strike Prevention
- SOP-LV-012 — CX-12 Plus – Kantech KT Integration

REMINDER:

Code text is periodically amended. This SOP reflects Article 3.4.6.16 as currently published; confirm against the current Ontario Building Code before relying on this document for a specific installation, and treat any conflict between this SOP and the current code text as resolved in favor of the code.